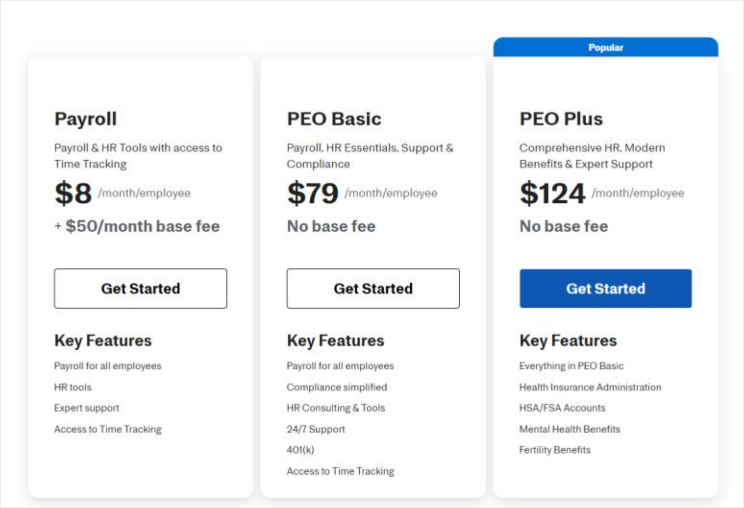
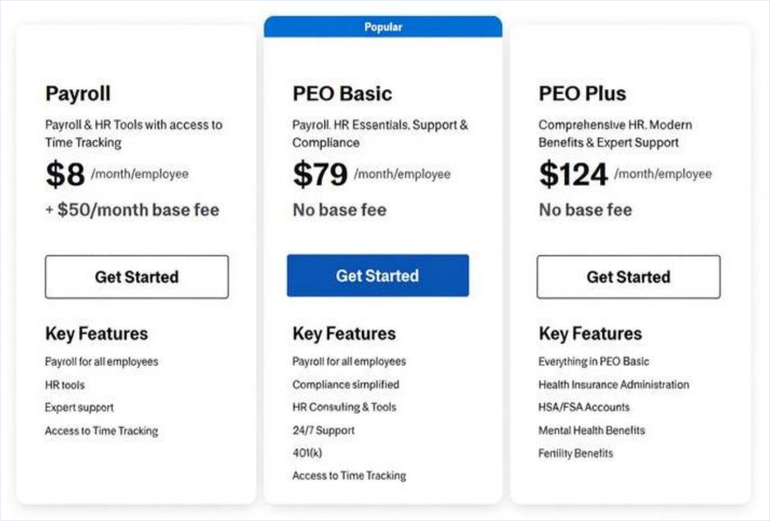
A/B Testing - Demo

Does changing the “Popular” banner to PEO basic increase conversion rates on Desktop?

Control – Group A



Treatment – Group B



Using Simulated Data

Shawn D’Souza

The Experiment

Hypothesis: Changing the popular banner to PEO basic does not have a difference in conversion Rates (CVR)

CVR: Total Conversions/Total views on Desktop.

CONTROL — Group A

Original website version

Users experienced the existing design and flow without any modifications. Served as the baseline for comparison.

n = 200 users

VS

TREATMENT — Group B

New website version

Users experienced the updated website design and flow. Changes were intended to improve the core user metric by enhancing the experience.

n = 200 users

Assignment unit: User | Both groups equally sized at 200 users each | Two-tailed hypothesis test

What We Found

+1.42

Metric Lift (Absolute)

Group B vs Group A

$p < 0.0001$

Statistical Significance

✓ Highly Significant ($\alpha = 0.05$)

+47.4%

Relative Improvement

Treatment over Control

Group B (new website version) outperformed Group A by 47.4% on the primary metric with overwhelming statistical confidence — both parametric and non-parametric tests confirm the result.

RECOMMENDATION: SHIP

Primary Metric: Group A vs Group B

3.004

Group A (Control)

Mean · std = 0.970 · n = 200

+1.42 (+47.4%)

4.428

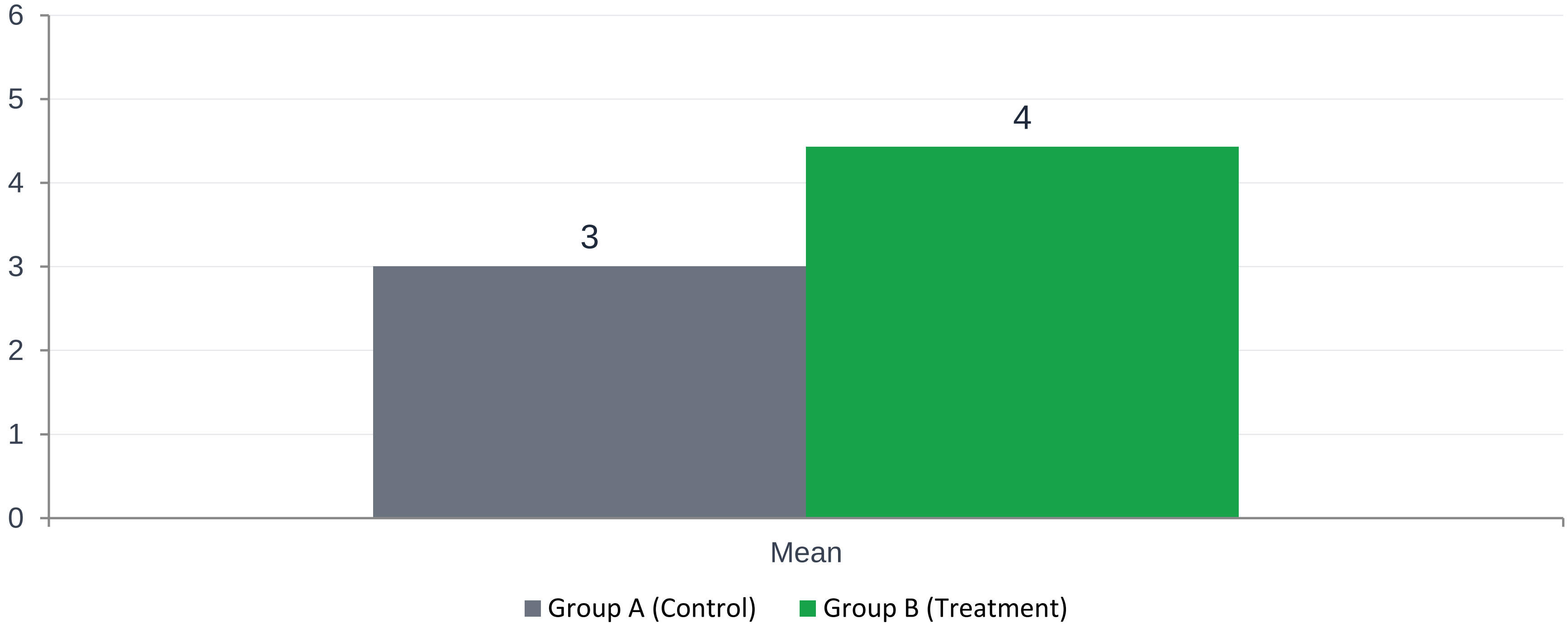
Group B (Treatment)

Mean · std = 0.968 · n = 200

Statistically Significant — Welch t-test: $t = -14.69$, $p < 0.0001$ | Mann-Whitney U: $p < 0.0001$ | Cohen's d = 1.47 (Large effect)

95% Confidence Interval for difference: [+1.23, +1.61] | Both groups passed Shapiro-Wilk normality test

Metric Distribution Comparison



Group B mean is 47.4% higher than Group A — a large, statistically significant difference ($p < 0.0001$, Cohen's $d = 1.47$)

Statistical Design & Robustness

Parameter	Value
Test used	Welch two-sample t-test (two-tailed)
Sample sizes	n = 200 per group (400 total)
Significance level α	0.05
t-statistic	−14.69
p-value	< 0.0001
95% CI (B – A)	[+1.23, +1.61]

Key Takeaways

- Both groups are normally distributed — parametric t-test is valid
- The entire 95% CI is positive — no scenario where A beats B
- Result is robust: p < 0.0001 vs threshold of 0.05

Recommendation

✓ SHIP IT

- Primary metric: Group B outperformed Group A by +1.42 (+47.4%) — $p < 0.0001$, large effect (Cohen's $d = 1.47$)
- Statistical robustness: Both parametric and non-parametric tests agree; 95% CI is entirely positive
- Sample adequacy: 200 users per group with both groups normally distributed — analysis is valid

Next Steps

- Roll out new website version (Group B) to 100% of users
- Monitor primary metric and guardrail KPIs for 30 days post-launch
- Document learnings and define hypothesis for the next experiment

Appendix

Methodology & Statistical Detail

Appendix A — Full Descriptive Statistics

Statistic	Group A (Control)	Group B (Treatment)	Difference (B – A)
Sample size (n)	200	200	—
Mean	3.004	4.428	+1.424 (+47.4%)
Std deviation	0.970	0.968	−0.002
Median	2.955	4.422	+1.467
Min	0.792	1.123	—
Max	6.189	7.270	—
Shapiro-Wilk p	0.398 ✓ Normal	0.497 ✓ Normal	—

** Min/Max are approximate values from the raw data files.*

Appendix B — Hypothesis & Statistical Design

"By deploying the new website version (Group B), we expect the primary user metric to increase significantly compared to the existing version (Group A)."

Parameter	Value
Significance level (α)	0.05 (5% false positive rate)
Statistical power (1−β)	0.80 (80% — standard for A/B tests)
Test type	Welch two-sample t-test (two-tailed) + Mann-Whitney U backup
Assignment unit	User (individual-level randomisation)
Sample size per variant	200 users
Observed effect size	Cohen's d = 1.47 (Large — threshold is 0.8)
Minimum runtime	Data collected from both groups simultaneously